

5 Partial Differentiation

1. Calculate the partial derivatives f_x and f_y of each of the following functions.

(a) $f(x, y) = (2x + 2y)^{10}$

$$f_x = 10 (2x + 2y)^9 \frac{\partial}{\partial x} (2x + 2y) = 20 (2x + 2y)^9$$

$$f_y = 10 (2x + 2y)^9 \frac{\partial}{\partial y} (2x + 2y) = 20 (2x + 2y)^9$$

(b) $f(x, y) = \ln(x + y)$

$$f_x = \frac{1}{x + y} \frac{\partial}{\partial x} (x + y) = \frac{1}{x + y}$$

$$f_y = \frac{1}{x + y} \frac{\partial}{\partial y} (x + y) = \frac{1}{x + y}$$

(c) $f(x, y) = x^y$

$$f_x = yx^{y-1}$$

$$f_y = x^y \ln x$$

(d) $f(x, y) = e^{x^2+y^2}$

$$f_x = e^{x^2+y^2} \frac{\partial}{\partial x} (x^2 + y^2) = 2x e^{x^2+y^2}$$

$$f_y = e^{x^2+y^2} \frac{\partial}{\partial y} (x^2 + y^2) = 2y e^{x^2+y^2}$$

(e) $f(x, y) = \frac{x - y}{x + y}$

$$f_x = \frac{(x + y) \frac{\partial}{\partial x} (x - y) - (x - y) \frac{\partial}{\partial x} (x + y)}{(x + y)^2} = \frac{(x + y) - (x - y)}{(x + y)^2} = \frac{2y}{(x + y)^2}$$

$$f_y = \frac{(x + y) \frac{\partial}{\partial y} (x - y) - (x - y) \frac{\partial}{\partial y} (x + y)}{(x + y)^2} = \frac{-(x + y) - (x - y)}{(x + y)^2} = \frac{-2x}{(x + y)^2}$$

(f) $f(x, y) = (x + y)(xy - 1) = x^2y - x + xy^2 - y$

$$f_x = 2xy - 1 + y^2$$

$$f_y = x^2 + 2xy - 1$$

$$(g) \quad f(x, y) = \sin^2(x - 3y)$$

$$\begin{aligned} f_x &= 2 \sin(x - 3y) \frac{\partial}{\partial x} (\sin(x - 3y)) \\ &= 2 \sin(x - 3y) \cos(x - 3y) \frac{\partial}{\partial x} (x - 3y) = \sin 2(x - 3y) \\ f_y &= 2 \sin(x - 3y) \frac{\partial}{\partial y} (\sin(x - 3y)) \\ &= 2 \sin(x - 3y) \cos(x - 3y) \frac{\partial}{\partial y} (x - 3y) = -3 \sin 2(x - 3y) \end{aligned}$$

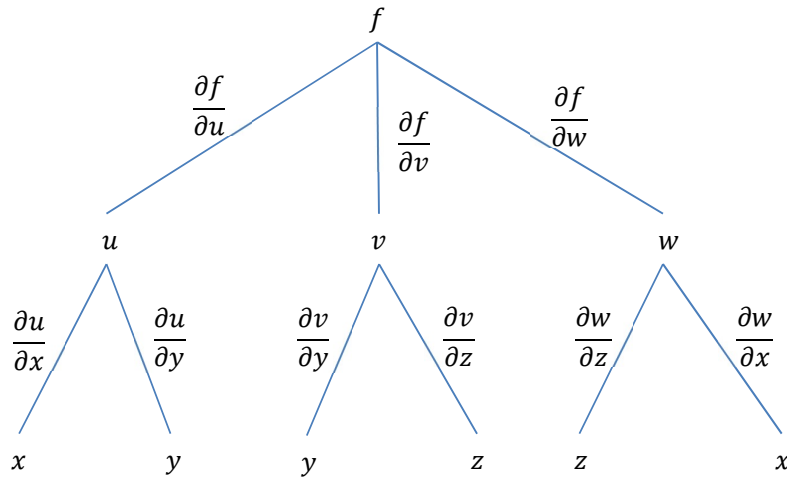
$$(h) \quad f(x, y) = e^{-x} \sin(x + y)$$

$$\begin{aligned} f_x &= e^{-x} \frac{\partial}{\partial x} (\sin(x + y)) + \sin(x + y) \frac{\partial}{\partial x} (e^{-x}) \\ &= e^{-x} \cos(x + y) \frac{\partial}{\partial x} (x + y) + \sin(x + y) e^{-x} \frac{\partial}{\partial x} (-x) \\ &= e^{-x} \cos(x + y) - \sin(x + y) e^{-x} = e^{-x} (\cos(x + y) - \sin(x + y)) \\ f_y &= e^{-x} \frac{\partial}{\partial y} (\sin(x + y)) = e^{-x} \cos(x + y) \frac{\partial}{\partial y} (x + y) = e^{-x} \cos(x + y) \end{aligned}$$

2. Assuming all partial derivatives exist, use the chain rule to prove each of the following.

(a) Given $f(u, v, w)$, where $u = x - y$, $v = y - z$, and $w = z - x$, show that

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0.$$



$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial v}{\partial y} = 1$$

$$\frac{\partial w}{\partial z} = 1$$

$$\frac{\partial u}{\partial y} = -1$$

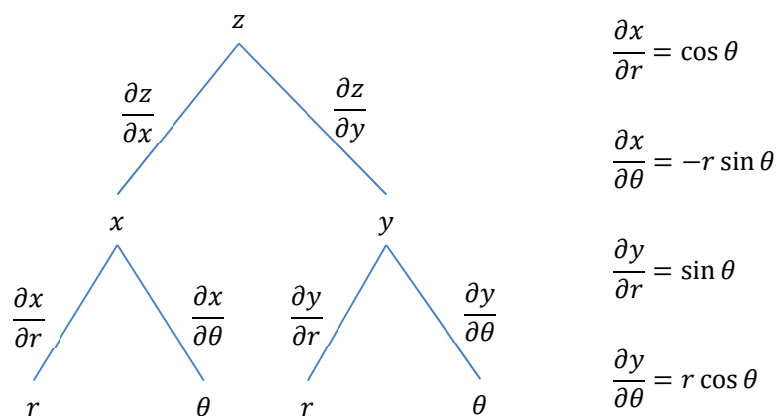
$$\frac{\partial v}{\partial z} = -1$$

$$\frac{\partial w}{\partial x} = -1$$

$$\begin{aligned} \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} &= \left(\frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial w} \frac{\partial w}{\partial x} \right) + \left(\frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} \right) + \left(\frac{\partial f}{\partial v} \frac{\partial v}{\partial z} + \frac{\partial f}{\partial w} \frac{\partial w}{\partial z} \right) \\ &= \left(\frac{\partial f}{\partial u} - \frac{\partial f}{\partial w} \right) + \left(-\frac{\partial f}{\partial u} + \frac{\partial f}{\partial v} \right) + \left(-\frac{\partial f}{\partial v} + \frac{\partial f}{\partial w} \right) = 0 \end{aligned}$$

(b) Given $z = f(x, y)$, where $x = r \cos \theta$ and $y = r \sin \theta$, show that

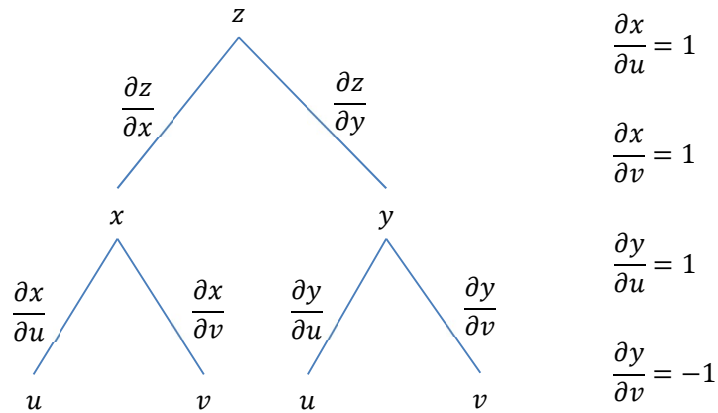
$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2.$$



$$\begin{aligned}
 \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 &= \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta}\right)^2 \\
 &= \left(\frac{\partial z}{\partial x} \cos \theta + \frac{\partial z}{\partial y} \sin \theta\right)^2 + \frac{1}{r^2} \left(-r \frac{\partial z}{\partial x} \sin \theta + r \frac{\partial z}{\partial y} \cos \theta\right)^2 \\
 &= \left(\frac{\partial z}{\partial x}\right)^2 \cos^2 \theta + 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \sin \theta \cos \theta + \left(\frac{\partial z}{\partial y}\right)^2 \sin^2 \theta \\
 &\quad + \left(\frac{\partial z}{\partial x}\right)^2 \sin^2 \theta - 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \sin \theta \cos \theta + \left(\frac{\partial z}{\partial y}\right)^2 \cos^2 \theta \\
 &= \left(\frac{\partial z}{\partial x}\right)^2 (\cos^2 \theta + \sin^2 \theta) + \left(\frac{\partial z}{\partial y}\right)^2 (\sin^2 \theta + \cos^2 \theta) \\
 &= \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2
 \end{aligned}$$

- (c) Given $z = f(x, y)$, where $x = u + v$ and $y = u - v$, show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial u} \frac{\partial z}{\partial v}.$$



$$\frac{\partial z}{\partial u} \frac{\partial z}{\partial v} = \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}\right) \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}\right) = \left(\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\right) \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = \left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2$$



- (d) If $z = f(x + ay) + g(x - ay)$, show that $\frac{\partial^2 z}{\partial y^2} = a^2 \frac{\partial^2 z}{\partial x^2}$.

Let $u = x + ay$ and $v = x - ay$. Then

$$\frac{\partial u}{\partial x} = 1 \quad \frac{\partial u}{\partial y} = a \quad \frac{\partial v}{\partial x} = 1 \quad \frac{\partial v}{\partial y} = -a$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = a \frac{\partial z}{\partial u} - a \frac{\partial z}{\partial v} = a \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right)$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = a \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) = a \left(\frac{\partial^2 z}{\partial u^2} \frac{\partial u}{\partial y} - \frac{\partial^2 z}{\partial v^2} \frac{\partial v}{\partial y} \right) = a^2 \left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \right)$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = \frac{\partial^2 z}{\partial u^2} \frac{\partial u}{\partial x} + \frac{\partial^2 z}{\partial v^2} \frac{\partial v}{\partial x} = \frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$$

$$\Rightarrow a^2 \frac{\partial^2 z}{\partial x^2} = a^2 \left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \right) = \frac{\partial^2 z}{\partial y^2}$$

3. Verify each of the following (k is a non-zero constant).

(a) $f(x, t) = e^{-t} \cos \frac{x}{k}$ satisfies the **heat equation** $\frac{\partial f}{\partial t} = k^2 \frac{\partial^2 f}{\partial x^2}$.

$$\frac{\partial f}{\partial t} = \cos \frac{x}{k} \frac{\partial}{\partial t} (e^{-t}) = -e^{-t} \cos \frac{x}{k}$$

$$\frac{\partial f}{\partial x} = e^{-t} \frac{\partial}{\partial x} \left(\cos \frac{x}{k} \right) = -\frac{1}{k} e^{-t} \sin \frac{x}{k}$$

$$\frac{\partial^2 f}{\partial x^2} = -\frac{1}{k} e^{-t} \frac{\partial}{\partial x} \left(\sin \frac{x}{k} \right) = -\frac{1}{k^2} e^{-t} \cos \frac{x}{k}$$

$$\Rightarrow k^2 \frac{\partial^2 f}{\partial x^2} = k^2 \left(-\frac{1}{k^2} e^{-t} \cos \frac{x}{k} \right) = -e^{-t} \cos \frac{x}{k} = \frac{\partial f}{\partial t}$$

(b) $f(x, t) = \sin 3kt \sin 3x$ satisfies the **wave equation** $\frac{\partial^2 f}{\partial t^2} = k^2 \frac{\partial^2 f}{\partial x^2}$.

$$\frac{\partial f}{\partial t} = \sin 3x \frac{\partial}{\partial t} (\sin 3kt) = 3k \sin 3x \cos 3kt$$

$$\Rightarrow \frac{\partial^2 f}{\partial t^2} = 3k \sin 3x \frac{\partial}{\partial t} (\cos 3kt) = -9k^2 \sin 3x \sin 3kt$$

$$\frac{\partial f}{\partial x} = \sin 3kt \frac{\partial}{\partial x} (\sin 3x) = 3 \sin 3kt \cos 3x$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = 3 \sin 3kt \frac{\partial}{\partial x} (\cos 3x) = -9 \sin 3kt \sin 3x$$

$$\Rightarrow k^2 \frac{\partial^2 f}{\partial x^2} = -9k^2 \sin 3kt \sin 3x = \frac{\partial^2 f}{\partial t^2}$$

(c) $f(x, t) = \frac{e^{-x^2/(4kt)}}{\sqrt{t}}$ satisfies the **diffusion equation** $\frac{\partial f}{\partial t} = k \frac{\partial^2 f}{\partial x^2}$.

$$\begin{aligned}\frac{\partial f}{\partial t} &= \frac{\sqrt{t} \frac{\partial}{\partial t} (e^{-x^2/4kt}) - e^{-x^2/4kt} \frac{\partial}{\partial t} (\sqrt{t})}{t} \\ &= \frac{\sqrt{t} e^{-x^2/4kt} \frac{\partial}{\partial t} \left(-\left(\frac{x^2}{4k}\right) t^{-1} \right) - e^{-x^2/4kt} \left(\frac{1}{2\sqrt{t}} \right)}{t} \\ &= \frac{\sqrt{t} e^{-x^2/4kt} \left(\frac{x^2}{4kt^2} \right) - e^{-x^2/4kt} \left(\frac{1}{2\sqrt{t}} \right)}{t} = \frac{e^{-x^2/4kt}}{t\sqrt{t}} \left(\frac{x^2}{4kt} - \frac{1}{2} \right) \\ \frac{\partial f}{\partial x} &= \frac{e^{-x^2/4kt}}{\sqrt{t}} \frac{\partial}{\partial x} \left(-\frac{x^2}{4kt} \right) = -\frac{x}{2kt\sqrt{t}} e^{-x^2/4kt} \\ \Rightarrow \frac{\partial^2 f}{\partial x^2} &= -\frac{1}{2kt\sqrt{t}} \left(e^{-x^2/4kt} \frac{\partial}{\partial x} (x) + x \frac{\partial}{\partial x} (e^{-x^2/4kt}) \right) \\ &= -\frac{1}{2kt\sqrt{t}} \left(e^{-x^2/4kt} + x \left(-\frac{x}{2kt} e^{-x^2/4kt} \right) \right) \\ &= -\frac{e^{-x^2/4kt}}{2kt\sqrt{t}} \left(1 - \frac{x^2}{2kt} \right) = \frac{e^{-x^2/4kt}}{kt\sqrt{t}} \left(\frac{x^2}{4kt} - \frac{1}{2} \right) \\ \Rightarrow k \frac{\partial^2 f}{\partial x^2} &= \frac{e^{-x^2/4kt}}{t\sqrt{t}} \left(\frac{x^2}{4kt} - \frac{1}{2} \right) = \frac{\partial f}{\partial t}\end{aligned}$$

(d) The pair: $u(x, y) = e^{-x} \sin y$ and $v(x, y) = e^{-x} \cos y$ satisfies the **Cauchy-Riemann equation** $u_x = v_y$ and $u_y = -v_x$.

$$\begin{aligned}u_x &= -e^{-x} \sin y & u_y &= e^{-x} \cos y \\ v_x &= -e^{-x} \cos y = -u_y & v_y &= -e^{-x} \sin y = u_x\end{aligned}$$

4. Calculate the gradient of each of the following functions at the specified point.

(a) $f(x, y) = \ln(x^2 + y^2)$ at $(1, 1)$

$$f_x = \frac{1}{x^2 + y^2} \frac{\partial}{\partial x} (x^2 + y^2) = \frac{2x}{x^2 + y^2}$$

$$f_y = \frac{1}{x^2 + y^2} \frac{\partial}{\partial y} (x^2 + y^2) = \frac{2y}{x^2 + y^2}$$

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix} = \frac{2}{x^2 + y^2} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \nabla f(1, 1) = \frac{2}{1+1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(b) $f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x$ at $(1, 1, 1)$

$$f_x = 2x + \frac{z}{x}$$

$$f_y = 2y$$

$$f_z = -4z + \ln x$$

$$\nabla f = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} = \begin{bmatrix} 2x + \frac{z}{x} \\ 2y \\ -4z + \ln x \end{bmatrix} \Rightarrow \nabla f(1, 1, 1) = \begin{bmatrix} 2 + 1 \\ 2 \\ -4 + \ln 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}$$

(c) $f(x, y, z) = 3e^x \cos(yz)$ at $(0, 0, 0)$

$$f_x = 3e^x \cos(yz)$$

$$f_y = -3ze^x \sin(yz)$$

$$f_z = -3ye^x \sin(yz)$$

$$\nabla f = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} = 3e^x \begin{bmatrix} \cos(yz) \\ -z \sin(yz) \\ -y \sin(yz) \end{bmatrix} \Rightarrow \nabla f(0, 0, 0) = 3e^0 \begin{bmatrix} \cos 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

(d) $f(x, y, z) = (x^2 + y^2 + z)^{-1/2} + \ln(xyz)$ at $(-1, 2, -1)$

$$f_x = \frac{-x}{(x^2 + y^2 + z)^{3/2}} + \frac{1}{x}$$

$$f_y = \frac{-y}{(x^2 + y^2 + z)^{3/2}} + \frac{1}{y}$$

$$f_z = \frac{-1/2}{(x^2 + y^2 + z)^{3/2}} + \frac{1}{z}$$

$$\nabla f = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} = \begin{bmatrix} 1/x - x(x^2 + y^2 + z)^{-3/2} \\ 1/y - y(x^2 + y^2 + z)^{-3/2} \\ 1/z - 0.5/(x^2 + y^2 + z)^{-3/2} \end{bmatrix}$$

$$\Rightarrow \nabla f(-1, 2, -1) = \begin{bmatrix} -1 + 1(1 + 4 - 1)^{-3/2} \\ 0.5 - 2(1 + 4 - 1)^{-3/2} \\ -1 - 0.5(1 + 4 - 1)^{-3/2} \end{bmatrix} = \begin{bmatrix} -0.875 \\ 0.25 \\ -1.0625 \end{bmatrix}$$

5. Construct the standard equation of the tangent plane and symmetric equation for the normal line to each of the following surfaces at the specified point on the surface.

(a) $z = \ln(x^2 + y^2)$ at $(1, 0, 0)$

$$z_x = \frac{2x}{x^2 + y^2}$$

$$z_y = \frac{2y}{x^2 + y^2}$$

$$\mathbf{N} = \begin{bmatrix} z_x(1, 0) \\ z_y(1, 0) \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$\text{Tangent plane: } \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x-1 \\ y-0 \\ z-0 \end{bmatrix} = 0 \Rightarrow 2(x-1) - (z-0) = 0 \Rightarrow 2x - 2 - z = 0$$

$$\text{Normal line: } \frac{x-1}{2} = \frac{z}{-1} \text{ and } y = 0$$

(b) $z = e^{-(x^2+y^2)}$ at $(0, 0, 1)$

$$z_x = -2xe^{-(x^2+y^2)}$$

$$z_y = -2ye^{-(x^2+y^2)}$$

$$\mathbf{N} = \begin{bmatrix} z_x(0, 0) \\ z_y(0, 0) \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$\text{Tangent plane: } \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x-0 \\ y-0 \\ z-1 \end{bmatrix} = 0 \Rightarrow -(z-1) = 0 \Rightarrow z-1 = 0$$

$$\text{Normal line: } x = 0, y = 0 \text{ and } z \in \mathbb{R}$$

(c) $z = 4x^2 + y^2$ at $(1, 1, 5)$

$$z_x = 8x$$

$$z_y = 2y$$

$$\mathbf{N} = \begin{bmatrix} z_x(1, 1) \\ z_y(1, 1) \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ -1 \end{bmatrix}$$

$$\text{Tangent plane: } \begin{bmatrix} 8 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x-1 \\ y-1 \\ z-5 \end{bmatrix} = 0 \Rightarrow 8(x-1) + 2(y-1) - (z-5) = 0$$

$$\Rightarrow 8x + 2y - z - 5 = 0$$

$$\text{Normal line: } \frac{x-1}{8} = \frac{y-1}{2} = \frac{z-5}{-1}$$

(d) $x^2 + y^2 + z^2 = 3$ at $(1, 1, 1)$

$$F(x, y, z) = x^2 + y^2 + z^2 - 3$$

$$F_x = 2x$$

$$F_y = 2y$$

$$F_z = 2z$$

$$\mathbf{N} = \nabla F(1, 1, 1) = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

$$\begin{aligned} \text{Tangent plane: } \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} x-1 \\ y-1 \\ z-1 \end{bmatrix} &= 0 \Rightarrow 2(x-1) + 2(y-1) + 2(z-1) = 0 \\ &\Rightarrow x + y + z - 3 = 0 \end{aligned}$$

$$\text{Normal line: } \frac{x-1}{2} = \frac{y-1}{2} = \frac{z-1}{2}$$

(e) $x^2 - 2y^2 + z^2 + yz = 2$ at $(2, 1, -1)$

$$F(x, y, z) = x^2 - 2y^2 + z^2 + yz - 2$$

$$F_x = 2x$$

$$F_y = -4y + z$$

$$F_z = 2z + y$$

$$\mathbf{N} = \nabla F(2, 1, -1) = \begin{bmatrix} 4 \\ -4-1 \\ -2+1 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \\ -1 \end{bmatrix}$$

$$\begin{aligned} \text{Tangent plane: } \begin{bmatrix} 4 \\ -5 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x-2 \\ y-1 \\ z+1 \end{bmatrix} &= 0 \Rightarrow 4(x-2) - 5(y-1) - (z+1) = 0 \\ &\Rightarrow 4x - 5y - z - 4 = 0 \end{aligned}$$

$$\text{Normal line: } \frac{x-2}{4} = \frac{y-1}{-5} = \frac{z+1}{-1}$$

(f) $yz = \ln(x+z)$ at $(0, 0, 1)$

$$F(x, y, z) = yz - \ln(x+z)$$

$$F_x = -\frac{1}{x+z}$$

$$F_y = z$$

$$F_z = y - \frac{1}{x+z}$$

$$\mathbf{N} = \nabla F(0, 0, 1) = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\begin{aligned} \text{Tangent plane: } \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x-0 \\ y-0 \\ z-1 \end{bmatrix} &= 0 \Rightarrow -x + y - (z-1) = 0 \Rightarrow -x + y - z + 1 = 0 \end{aligned}$$

$$\text{Normal line: } \frac{x}{-1} = \frac{y}{1} = \frac{z-1}{-1}$$

6. Find the relative extrema and saddle points of each of the following functions.

(a) $f(x, y) = x^2 + y^2 + x^2y + 4$

$$f_x = 2x + 2xy$$

$$f_y = 2y + x^2$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} x + xy = 0 \\ 2y + x^2 = 0 \end{cases}$$

The critical points are at $(0, 0), (\sqrt{2}, -1), (-\sqrt{2}, -1)$.

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(2x + 2xy) & \frac{\partial}{\partial y}(2x + 2xy) \\ \frac{\partial}{\partial x}(2y + x^2) & \frac{\partial}{\partial y}(2y + x^2) \end{vmatrix} = \begin{vmatrix} 2 + 2y & 2x \\ 2x & 2 \end{vmatrix} = 4 + 4y - 4x^2$$

At $(0, 0)$: $D_f(0, 0) = 4 > 0$ and $f_{xx}(0, 0) = 2 > 0 \Rightarrow (0, 0, 4)$ is a rel. min. point

At $(\sqrt{2}, -1)$: $D_f(\sqrt{2}, -1) = -8 < 0 \Rightarrow (\sqrt{2}, -1, 5)$ is a saddle point

At $(-\sqrt{2}, -1)$: $D_f(-\sqrt{2}, -1) = -8 < 0 \Rightarrow (-\sqrt{2}, -1, 5)$ is a saddle point

(b) $f(x, y) = e^{4y-x^2-y^2}$

$$f_x = -2xe^{4y-x^2-y^2}$$

$$f_y = (4 - 2y)e^{4y-x^2-y^2}$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} -2x = 0 \\ 4 - 2y = 0 \end{cases}$$

The critical point is at $(0, 2)$.

$$\begin{aligned} D_f &= \begin{vmatrix} \frac{\partial}{\partial x}(-2xe^{4y-x^2-y^2}) & \frac{\partial}{\partial y}(-2xe^{4y-x^2-y^2}) \\ \frac{\partial}{\partial x}((4-2y)e^{4y-x^2-y^2}) & \frac{\partial}{\partial y}((4-2y)e^{4y-x^2-y^2}) \end{vmatrix} \\ &= 4e^{2(4y-x^2-y^2)} \begin{vmatrix} 2x^2 - 1 & x(2y - 4) \\ x(2y - 4) & 2(2 - y)^2 - 1 \end{vmatrix} \\ &= 4e^{2(4y-x^2-y^2)}((2x^2 - 1)(2(2 - y)^2 - 1) - x^2(2y - 4)^2) \end{aligned}$$

At $(0, 2)$: $D_f(0, 2) = 4e^8(1 - 0) > 0$ and $f_{xx}(0, 2) = -2e^4 < 0$

$\Rightarrow (0, 2, e^4)$ is a rel. max. point

(c) $f(x, y) = 2x^3 + 2y^3 - 9x^2 + 3y^2 - 12y$

$$f_x = 6x^2 - 18x$$

$$f_y = 6y^2 + 6y - 12$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} x^2 - 3x = 0 \\ y^2 + y - 2 = 0 \end{cases}$$

The critical points are at $(0, 1), (0, -2), (3, 1), (3, -2)$.

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(6x^2 - 18x) & \frac{\partial}{\partial y}(6x^2 - 18x) \\ \frac{\partial}{\partial x}(6y^2 + 6y - 12) & \frac{\partial}{\partial y}(6y^2 + 6y - 12) \end{vmatrix} = \begin{vmatrix} 12x - 18 & 0 \\ 0 & 12y + 6 \end{vmatrix}$$

$$= 36(2x - 3)(2y + 1)$$

At $(0, 1)$: $D_f(0, 1) = 36(-3)(3) < 0 \Rightarrow (0, 1, -7)$ is a saddle point

At $(0, -2)$: $D_f(0, -2) = 36(-3)(-3) > 0$ and $f_{xx}(0, -2) = -18 < 0$
 $\Rightarrow (0, -2, 20)$ is a rel. max. point

At $(3, 1)$: $D_f(3, 1) = 36(3)(3) > 0$ and $f_{xx}(3, 1) = 18 > 0$
 $\Rightarrow (3, 1, -34)$ is a rel. min. point

At $(3, -2)$: $D_f(3, -2) = 36(3)(-3) < 0 \Rightarrow (3, -2, -7)$ is a saddle point

(d) $f(x, y) = x^4 + y^4 + 4xy$

$$f_x = 4x^3 + 4y$$

$$f_y = 4y^3 + 4x$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} x^3 + y = 0 \\ y^3 + x = 0 \end{cases}$$

The critical points are at $(0, 0), (1, -1), (-1, 1)$.

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(4x^3 + 4y) & \frac{\partial}{\partial y}(4x^3 + 4y) \\ \frac{\partial}{\partial x}(4y^3 + 4x) & \frac{\partial}{\partial y}(4y^3 + 4x) \end{vmatrix} = \begin{vmatrix} 12x^2 & 4 \\ 4 & 12y^2 \end{vmatrix} = 16(9x^2y^2 - 1)$$

At $(0, 0)$: $D_f(0, 0) = -16 < 0 \Rightarrow (0, 0, 0)$ is a saddle point

At $(1, -1)$: $D_f(1, -1) = 128 > 0$ and $f_{xx}(1, -1) = 12 > 0$
 $\Rightarrow (1, -1, -2)$ is a rel. min. point

At $(-1, 1)$: $D_f(-1, 1) = 128 > 0$ and $f_{xx}(-1, 1) = 12 > 0$
 $\Rightarrow (-1, 1, -2)$ is a rel. min. point



(e) $f(x, y) = x^2y^4$

$$f_x = 2xy^4$$

$$f_y = 4x^2y^3$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} xy^4 = 0 \\ x^2y^3 = 0 \end{cases}$$

The critical points are at $(t, 0), (0, t)$ for all real t .

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(2xy^4) & \frac{\partial}{\partial y}(2xy^4) \\ \frac{\partial}{\partial x}(4x^2y^3) & \frac{\partial}{\partial y}(4x^2y^3) \end{vmatrix} = \begin{vmatrix} 2y^4 & 8xy^3 \\ 8xy^3 & 12x^2y^2 \end{vmatrix} = 8(3x^2y^6 - 8x^2y^6) = -40x^2y^6$$

Note that $D_f = 0$ at all critical points.

However, since $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$, and $f(t, 0) = f(0, t) = 0$ for all $t \in \mathbb{R}$, we conclude $(t, 0, 0)$ and $(0, t, 0)$ are relative minimum points for all $t \in \mathbb{R}$.



(f) $f(x, y) = x^2y^3$

$$f_x = 2xy^3$$

$$f_y = 3x^2y^2$$

$$f_x = 0 \text{ and } f_y = 0 \Rightarrow \begin{cases} xy^3 = 0 \\ x^2y^2 = 0 \end{cases}$$

The critical points are $(t, 0), (0, t)$ for all real t .

$$D_f = \begin{vmatrix} \frac{\partial}{\partial x}(2xy^3) & \frac{\partial}{\partial y}(2xy^3) \\ \frac{\partial}{\partial x}(3x^2y^2) & \frac{\partial}{\partial y}(3x^2y^2) \end{vmatrix} = \begin{vmatrix} 2y^3 & 6xy^2 \\ 6xy^2 & 6x^2y \end{vmatrix} = 12(x^2y^4 - 3x^2y^4) = -24x^2y^4$$

Note that $D_f = 0$ at all critical points.

Since $y > 0 \Rightarrow f(x, y) > 0$ and $y < 0 \Rightarrow f(x, y) < 0$ for any real x , we conclude that

$(t, 0, 0)$ are saddle points for all $t \in \mathbb{R}$,

$(0, t, 0)$ are relative minimum points for all $t > 0$,

$(0, t, 0)$ are relative maximum points for all $t < 0$.

7. Suppose a three-variable function $F(x, y, z)$ has continuous first and second partial derivatives. The **Laplacian operator** ∇^2 on F is defined as

$$\nabla^2 F = \nabla \cdot \nabla F = F_{xx} + F_{yy} + F_{zz}$$

A three-variable function F is called **harmonic** in the domain D if $\nabla^2 F = 0$ in D . Determine whether each of the following functions is harmonic on its domain.

(a) $F = e^{-x} \sin y$

$$F_x = -e^{-x} \sin y \Rightarrow F_{xx} = e^{-x} \sin y$$

$$F_y = e^{-x} \cos y \Rightarrow F_{yy} = -e^{-x} \sin y$$

$$F_z = 0 \Rightarrow F_{zz} = 0$$

$$\nabla^2 F = F_{xx} + F_{yy} + F_{zz} = e^{-x} \sin y - e^{-x} \sin y + 0 = 0 \Rightarrow F \text{ is harmonic}$$

(b) $F = xyz$

$$F_x = yz \Rightarrow F_{xx} = 0$$

$$F_y = xz \Rightarrow F_{yy} = 0$$

$$F_z = xy \Rightarrow F_{zz} = 0$$

$$\nabla^2 F = F_{xx} + F_{yy} + F_{zz} = 0 \Rightarrow F \text{ is harmonic}$$

(c) $F = e^{-x}(\cos y - \sin y)$

$$F_x = -e^{-x}(\cos y - \sin y) \Rightarrow F_{xx} = e^{-x}(\cos y - \sin y)$$

$$F_y = e^{-x}(-\sin y - \cos y) \Rightarrow F_{yy} = e^{-x}(\cos y - \sin y)$$

$$F_z = 0 \Rightarrow F_{zz} = 0$$

$$\nabla^2 F = F_{xx} + F_{yy} + F_{zz} = e^{-x}(\cos y - \sin y) + e^{-x}(\cos y - \sin y) + 0 = 0$$

Thus, F is harmonic.

(d) $F = (x^2 + y^2 + z^2)^{-1/2}$

$$F_x = -x(x^2 + y^2 + z^2)^{-3/2} \Rightarrow F_{xx} = (2x^2 - y^2 - z^2)(x^2 + y^2 + z^2)^{-5/2}$$

$$F_y = -y(x^2 + y^2 + z^2)^{-3/2} \Rightarrow F_{yy} = (-x^2 + 2y^2 - z^2)(x^2 + y^2 + z^2)^{-5/2}$$

$$F_z = -z(x^2 + y^2 + z^2)^{-3/2} \Rightarrow F_{zz} = (-x^2 - y^2 + 2z^2)(x^2 + y^2 + z^2)^{-5/2}$$

$$\nabla^2 F = \frac{(2x^2 - y^2 - z^2) + (-x^2 + 2y^2 - z^2) + (-x^2 - y^2 + 2z^2)}{(x^2 + y^2 + z^2)^{5/2}} = 0$$

Thus, F is harmonic.